

INTRO

Radio telescopes have long served an integral role in Radio Astronomy. Radio telescopes go beyond the human eye and the spectrum of visible light in order to view wavelengths necessary for the study of the chemical composition of celestial objects, the speed of celestial objects, and were instrumental in the discovery of dark matter. However, in recent years, the growth of WiFi services, Bluetooth connections, and relay transmissions in electronic devices has generated false readings in radio telescopes and has inhibited astronomers' abilities to correctly identify and classify celestial objects. As noted by Thompson (2014), "technical advances in electromagnetics, signal processing and processing power have led to a significant increase in sensitivity... with this increase, radio frequency interference (RFI) has become a much larger problem." Given that professionals are using more expensive setups, it is even more pressing for those not in possession of affluence or a laboratory. Lower-cost radio telescopes do not contain nearly as much clarity in their antennae, and cannot differentiate in spectrums at further distances very reliably. At the moment, the literature has produced multiple solutions to the issue. Some of the solutions involve mitigation as data is being taken in; however, such methods slow down and halt data unless done in more expensive areas. Other methods involve utilizing an adaptive filter: a separate antenna of much lower quality than the one utilized. From there, the readings from the lower-quality antenna are subtracted. However, this method requires the purchase of another antenna, and that too, one of high enough quality to gauge the RFI in the entirety of one's area. Barnbaum and Bradley (1998) explain that this "reference antenna" must be designed to receive "only the RFI," while the primary antenna receives both the signal and the RFI. Of these 2 main methods, the adaptive filter may present itself to be much more cost-effective; however, it is still out of reach of the typical backyard astronomer due to its cost and complex setup. Given, there is

one method not yet implemented in radio telescopes, but rather in WiFi receivers known as Doppler-assisted Wireless Communication or DAWC, which is unlike the other methods in the way that it is preventative, rather than adjusting real-time afterwards, it "shifts the interfering signals away from the desired signal band" before the program runs (Basnayaka & Ratnarajah, 2019). Noting DAWC as a potential solution for RFI in backyard astronomers' setups still introduces many questions about the process itself, such as the cost, the size of the telescope, and whether or not the process works on a radio astronomy scale. However, despite the unknowns, it could provide a valuable resource for backyard astronomers and professionals alike for saving money, and should therefore be pursued further.

HISTORICAL CONTEXT AND LITERATURE REVIEW

Looking into the past, there have been many attempts to mitigate RFI. Some methods acted upon the telescope itself, and others upon external factors. Starting in 1998, Barnbaum et al. set out to solve RFI through adaptive cancellation. In this process, Barnbaum et al. utilized two antennae, where one antenna was much weaker than the other. They did this because a weaker antenna would capture less at further distances, and therefore only the closer noise, which is bound to be RFI. From here, Barnbaum et al. can use a process known as interference excision to cut out the signal from the weaker antenna to cut out RFI. However, this process has drawbacks: the weaker antenna still needs to be marginally strong. The larger antenna also needs to be marginally stronger than the weaker antenna. The process also is not very successful in terms of fully removing signals, as the second antenna needs to be perfectly tuned to get an accurate signal. Barnbaum and Bradley (1998) noted that "even a small amount of signal in the reference channel will result in a cancellation of the signal in the primary channel." These facts combined mean

that a consumer of radio telescopes would need to purchase one telescope for the price of 2 antennae. For a backyard astronomer, this is far too expensive.

In the years following, researchers shied away from adaptive cancellation. In 2001, Fridman and Baan (2001) looked into the uses of different algorithms for antennae. Some algorithms came up, such as temporal domain rejection, frequency rejection, and spatial filtering. Each method selectively blocks out certain frequencies that correlate with RFI; however, blocking out specific frequencies leads to a loss of detail, and the use of algorithms results in slower application times. As they summarize, "the suppression of an RFI signal achieved with a certain method depends on the fractional intensity of the RFI signal... and its spatial, temporal and spectral characteristics" (Fridman & Baan, 2001). On larger-scale systems, algorithms may not provide such a large issue, but in backyard astronomer setups, any form of lag can delay and regress data. A backyard astronomer also would not be able to access higher-end antennae, meaning that the antennae already do not split frequencies optimally. Loss of data will have much more effect on a backyard astronomer than on a large-scale laboratory, making this method not entirely optimal for a backyard astronomer. Following this attempt, researchers revisited adaptive cancellation; Li et al. (2023) added functionality to Barnbaum's initial setup, rather than utilizing interference excision, he added the gain function. The gain function allowed for changes in the amount of the weaker antenna's signal taken in while the system runs, and therefore does not require perfect tuning of the weaker antenna. Li et al. (2023) state that this "adaptive filter algorithm flexibly adjusts the filter gain automatically, based on the correlation between the RFI reference signal and the observation signal." This issue resolves the need for a perfect choice of a secondary antenna, but it still requires 2 antennae with similarly large margins to Barnbaum et al.'s design, thus making it roughly the same price as Barnbaum's design, which, as concluded prior, is not

suitable for a backyard astronomer. Over time, more and more researchers have attempted solutions to the issue of RFI by changing the code surrounding the telescopes or changing the design of the telescope itself. Changes to the adaptive systems continue to this day, as well as novel algorithmic approaches to resolving the issue; however, with the way these methods are proceeding, they are less focused on lowering computational and/or instrumental costs and more focused on growing effectiveness. Although RFI interference is becoming less of a problem due to ongoing research, it is not becoming any more accessible to backyard astronomers through current physical and computational approaches.

While physical and computational modes seem the most intuitive for decreasing RFI, some methods have been put in place that do not require changes to the actual telescope itself. As Want S. et al. detail, there are proactive and reactive solutions to RFI. Reactive solutions are solutions that require computational and physical adjustments, but proactive solutions are those that can be done before the telescope is operated to enhance the performance of the radio telescope. Some methods detailed by Want S. et al. are to get a contract from the government to build in the area, or to choose a location outside of typical RFI zones. However, all of these methods are completely unrealistic for the average person. Not only is it impossible to have a restricted radio wave area in a local neighborhood, but reserving a specific place without radio interference is hard, especially for the average person. It is a similar situation for the method determined by Huang et al. (2020). In their experiment, they found that rotating a circuit board within an RFI emitting device could significantly reduce RFI at the correct orientation. They concluded that "rotating the source to reduce RFI is implemented in such a way that it does not compromise the signal integrity" (Huang et al., 2020). The research implies that laboratory environments could work to limit and control the orientations of their RFI emitting devices, however, much like the

prior example, it is hard for an average backyard astronomer to change any circuits in their house or around their neighbourhood such that RFI is mitigated. Sometimes, the inability for a backyard astronomer to perform proactive methods hinges on the accessibility of the tools. In the paper by Rao (1973), they introduce a method of locating RFI emitters utilizing satellites, utilizing the locations of RFI, they are able to both locate a position without much RFI, as well as hone their own antennae to changes in the RFI in that area. Rao (1973) discusses "using doppler techniques for determining the location of ground based emitters" which could help steer an antenna away from noise. However, this wouldn't be much use for a backyard radio astronomer as they do not necessarily possess technical knowledge of honing telescopes, and as discussed earlier, they do not have the ability to place a radio telescope wherever they want. Also, most backyard astronomers do not have the ability to put a satellite into space. Overall, considering the most prominent proactive methods introduced, the applicability of proactive RFI mitigation methods seems to be lower for backyard astronomers. Between the applicability of the reactive and proactive methods, backyard astronomers would most benefit from changes to the reactive methods.

It is understood that reactive methods work the best in comparison to proactive methods, but as concluded earlier, reactive methods are currently lacking in their progression towards affordability. Thus, it is necessary to find a method that integrates using a process applicable to a single antenna as opposed to two. To solve this problem the doppler shift is introduced. As per Basnayaka and Ratnarajah (2019), the Doppler effect "appears in radio frequency (RF) wireless communication when there exists a relative movement between the transmitter and the receiver." The doppler effect has long been used in radio astronomy, mostly for the purpose of accounting for changes in frequencies and wavelengths in radio waves, noted here is that the doppler effect

does provide an effect upon the telescope and frequencies detected due to the rotation of the earth. Additionally, it shifts frequencies back and forth in the radio telescopes output. It is possible to use this effect in such a way that RFI is shifted out of the spectrum. In revisiting Rao's (1973) paper on locating RFI sources, he was able to discern RFI utilizing the doppler effect using a satellite telescope. With this paper, Rao demonstrated that the doppler effect had an effect upon RFI, which may prove useful, especially due to it being usable on a single antenna. However, he did so on a satellite which is much more powerful than a backyard telescope. In Basnayaka and Ratnarajah's (2019) research, his team utilizes a process known as Doppler Assisted Wireless Communication (DAWC). In this process, a receiver is rotated about an origin whilst facing a target the entire time. This system "exploits Doppler effect and uses kinetic energy for co-channel interference (CCI) mitigation" (Basnayaka & Ratnarajah, 2019). This process has the effect of completely shifting the RFI out of the spectrum analysed. It also allows for the use of "one antenna/receiver," making it optimal for cost. However, it has only been implemented in WiFi communication devices, which are much smaller than radio telescopes. The speed and costs necessary to build such a device are unknown, and thus still in need of research. However, it is indisputable that the doppler effect approach is possible with radio telescopes.

It is concluded that the doppler effect approach is possible with radio telescopes, but this has not actually been implemented yet. To implement this in a method that aligns with the backyard astronomer, Reckziegel et al. (2022) provided a base design determined to be friendly for a backyard astronomer, specifically a "Small Radio Telescope for Observing the Neutral Hydrogen Line." This method includes both a receiver and stand. As per Basnayaka and Ratnarajah (2019), the rotating receiver "shifts the interfering signals away from the desired signal band." It would

make the most sense to keep the receiver of Reckziegel et al.'s system, and utilize it on a stand/base designed similarly to Basnayaka et al.'s. Reckzeigel's system uses an antenna as well as an SDR to take in antenna data, but the stand remain static, and does not rotate. In Basnayaka's system, the receiver is neither strong enough or heavy enough to fulfill the needs of a radio receiver. However, the rotational bottom is the basis of the doppler effect. It will be necessary to adjust the bottom system such that it can support weight and attain necessary speeds. Therefore, a basis for the design has been decided. However, it will be necessary to find a method to test it. To test the effectiveness of a device at preventing RFI, Thompson (2014) created a composite known as SNR. He explains that "the performance of an RFI mitigation algorithm is measured by its ability to reduce the RFI power while preserving the astronomical signal." The paper also evaluated the SNR of other known methods, this fact allows for the direct comparison with other known methods. Overall, through much documentation, there is enough research to conclude that a DAWC system for a radio telescope is possible.

Looking back through the literature, it is clear that there are many ways to mitigate RFI, both proactive and reactive. However, for a backyard astronomer, DAWC methods could provide the most affordable and plausible solution. While the current methods of RFI mitigation are sound, they are not affordable, and overall, may require much more economic, political, and social power than a backyard astronomer may have. At the current moment in radio astronomy, there are no solutions for backyard astronomers to join in on radio astronomy without major RFI concerns. The research following must focus on affordability, size constraints, effectiveness, and ease of use in order to fit the niche of backyard astronomers.